Passenger Assignment Model in a Public Transportation Network

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1 Introduction

The objective of this project is to exploit operational data provided by TPG (Geneva public transportation operator) in order to infer an aggregate demand representation. The demand is characterized by a flow between each origin–destination (OD) pair and each desired departure time interval.

The project follows four main steps:

1. Develop a detailed network representation of the public transportation system.
2. Formulate an assignment model that maps an OD matrix to passenger flows on the network.
3. Define measurement equations linking model outputs to observed data, yielding a likelihood function.
4. Use Bayesian inference (baseline implementation: NumPyro with variational inference) to approximate the posterior distribution of the OD matrix.

2 Modeling Elements

2.1 Spatial Representation

Let $\mathcal{S}$ denote the set of stops in the area of interest. Each stop can act as an origin, a destination, or a transfer location.

Definition 1 (OD pair). *An origin–destination pair is a tuple*

$$q = (s_o, s_d) \in \mathcal{Q} = \mathcal{S} \times \mathcal{S}.$$

2.2 Temporal Representation

The analysis period is partitioned into a set $\mathcal{T}$ of departure time intervals. Each interval represents a desired departure time window.

Definition 2 (Departure time interval). *Each time bin $t \in \mathcal{T}$ corresponds to a closed interval*

$$[a_t, b_t] \subset \mathbb{R}, \qquad a_t < b_t,$$

where a_t and b_t denote the start and end time (in minutes) of the desired departure interval.

For each departure interval $t \in \mathcal{T}$, the desired departure window is the interval $[a_t, b_t]$.

Definition 3 (OD demand). *For each OD pair* $q \in \mathcal{Q}$ *and departure time interval* $t \in \mathcal{T}$*, let*

$$f_q^t \geqslant 0$$

denote the number of passengers wishing to depart during interval t *from origin* s_o *to destination* s_d*.*

The full demand vector is denoted by

$$\mathbf{f} = \{f_q^t : q \in \mathcal{Q},\ t \in \mathcal{T}\}.$$

2.3 Baseline demand and deviation parameterization (Bayesian layer)

The model assumes that the analyst can provide an *a-priori* demand matrix $\mathbf{f}^{(0)} = \{f_q^{t,(0)}\}$, aligned with the OD/time-bin indexing of the scenario. Rather than placing a prior directly on the demand levels $\mathbf{f}$, we parameterize demand in terms of *log-deviations* from $\mathbf{f}^{(0)}$. This improves numerical stability, facilitates weakly informative priors, and makes the role of prior information explicit.

Modeling choice (log-deviation). For each OD/time-bin cell, introduce a latent deviation $\delta_q^t \in \mathbb{R}$ and define the demand used by the assignment as

$$f_q^t = f_q^{t,(0)} \exp(\delta_q^t).$$

This transformation guarantees $f_q^t \geqslant 0$ whenever $f_q^{t,(0)} \geqslant 0$ and preserves the baseline structure up to multiplicative adjustments.

Zero baseline cells. If $f_q^{t,(0)} = 0$, then $f_q^t = 0$ for all values of δ_q^t. Therefore, OD cells that are zero in the baseline matrix remain structurally zero during inference. In order to allow strictly positive deviations from zero, a small but non zero seed value must be provided: $f_q^{t,(0)} = \varepsilon > 0$.

Prior scale. The deviations are assumed independent and normally distributed:

$$\delta_q^t \sim \mathcal{N}(0, \sigma_z^2).$$

The parameter $\sigma_z > 0$ controls the strength of regularization. Small values of σ_z constrain demand to remain close to the baseline, while large values allow substantial multiplicative deviations. In the current implementation, σ_z is a fixed hyperparameter.

Separation of responsibilities. The assignment procedure is treated as a deterministic mapping

$$(\mathbf{f}, \theta) \longmapsto x(\mathbf{f}; \theta)$$

and remains agnostic to priors and baseline information. The construction of $\mathbf{f}$ from $\mathbf{f}^{(0)}$ and $\boldsymbol{\delta}$ is performed in the *Bayesian layer* (implemented using NumPyro and stochastic variational inference), which: (i) reads the baseline $\mathbf{f}^{(0)}$ provided by the user, (ii) draws deviations $\boldsymbol{\delta}$ from a prior distribution, and (iii) computes $\mathbf{f}$ using the expression above before invoking the assignment. This design avoids introducing additional indexing conventions and keeps the assignment module independent of inference choices.

Desired departure interval and admissible boarding window. For each OD pair $q \in \mathcal{Q}$ and departure time interval $t \in \mathcal{T}$, we associate the desired departure *interval* $[a_t, b_t]$ and an admissible schedule-deviation window of half-width $\Delta_{\max}^{\text{access}} > 0$ (user-defined, e.g. 15 minutes).

Admissible access links. An access link is included in the graph *only if* the departure time τ is not too far from the desired interval, namely

$$\tau \in \left[a_t - \Delta_{\max}^{\text{access}},\ b_t + \Delta_{\max}^{\text{access}}\right],$$

where τ denote the actual departure time (in minutes) of a boarding at the origin stop. As described later, the boarding is modeled by a link, that is included if the distance from τ to the interval $[a_t, b_t]$ does not exceed $\Delta_{\max}^{\text{access}}$.

Schedule-deviation penalty. When the access link is included, its generalized cost is defined by a (possibly asymmetric) piecewise-linear penalty with a *flat* (zero-penalty) region inside the interval:

$$c_{\text{access}}(\tau; t) = \beta_{\text{early}}\ \max\bigl(0,\ a_t - \tau\bigr) + \beta_{\text{late}}\ \max\bigl(0,\ \tau - b_t\bigr),$$

where $\beta_{\text{early}} \geqslant 0$ and $\beta_{\text{late}} \geqslant 0$ are user-defined coefficients. Therefore,

$$c_{\text{access}}(\tau; t) = 0 \quad \text{whenever} \quad \tau \in [a_t, b_t],$$

and penalties apply only to departures strictly earlier than a_t or strictly later than b_t.

Note that the thresholds $\Delta_{\max}^{\text{access}}$ and $\Delta_{\max}^{\text{transfer}}$ are fixed configuration parameters; the assignment is differentiable w.r.t. costs and OD flows given a fixed graph.

Implementation note (departure interval as OD/time-bin data). Although centroid-in nodes are not time-tagged in the graph, the desired departure *interval* $[a_t, b_t]$ remains an attribute of the OD/time-bin pair. In computations, the access-link penalty is evaluated by combining the interval bounds (a_t, b_t) with the clock time τ of the *departure event node* reached by that access link. This yields a zero penalty for departures within $[a_t, b_t]$ and an early/late penalty based on the nearest bound otherwise, without introducing backward-in-time links in the acyclic time-expanded network.

2.4 Network Representation

The public transportation system is represented as a directed time-expanded graph. All timetable times (arrival/departure event times) are expressed in minutes on the same time axis as the departure-time intervals.

A key modeling choice is (i) the separation of each stop centroid into two distinct nodes (an origin *centroid-in* and a destination *centroid-out*), and (ii) the split of each timetable event into an *arrival* and a *departure* node. This second split prevents ill-defined instantaneous moves (e.g., transferring or boarding at the exact same time as arrival) and ensures that all links respect a strict chronological (or lexicographic) order, so that the resulting time-expanded network is acyclic.

Definition 4 (Nodes)**.** *The node set consists of three families of nodes:*

- ***Centroid-in nodes (non time-tagged, indexed by departure interval):*** *for each stop* $s \in \mathcal{S}$ *and for each departure interval* $t \in \mathcal{T}$*, a node*

$$(s, t)^{\text{in}}$$

used as a demand-injection handle for passengers departing from stop s *with desired departure interval* t*. These nodes do not correspond to a physical time instant; they are placed* before all event nodes *in the topological order (conceptually at* $-\infty$*). The interval index* t *is used only to (i) select admissible boarding events through the access-window rule, and (ii) compute schedule-deviation penalties.*

- ***Centroid-out nodes (non time-tagged):*** *for each stop* $s \in \mathcal{S}$, *a node* s^{out} *representing the point where passengers arriving at stop* s *exit the network. These nodes are not associated with a physical time; they are placed* after all event nodes *in the topological order (conceptually at* $+\infty$*) so that the global graph remains acyclic.*

- ***Timetable event nodes (trip-specific, split into arrival and departure):*** *for each stop* s, *each scheduled arrival time* τ^{arr} *and departure time* τ^{dep} *associated with a given vehicle run (trip), we create two nodes*

$$(s, \tau^{\text{arr}})^{\text{arr}} \qquad \textit{and} \qquad (s, \tau^{\text{dep}})^{\text{dep}}.$$

  These nodes are trip-specific*: if two trips serve the same stop at the same clock time, they still correspond to different event nodes.*

Definition 5 (Lines and event-to-line mapping)**.** *Let* $\mathcal{L}$ *denote the set of public transport lines. Each timetable event node (arrival or departure) is associated with exactly one operated line, captured by the mapping*

$$\lambda : \mathcal{V}^{\text{event}} \to \mathcal{L}, \qquad \nu \mapsto \lambda(\nu),$$

where $\mathcal{V}^{\text{event}}$ *is the set of timetable event nodes. In practice,* $\lambda(\nu)$ *is derived from the timetable: event nodes created from a given trip inherit the line identifier of that trip. Therefore, for a given trip at stop* s, *the corresponding arrival and departure nodes share the same line label.*

Centroid-in nodes are not time-tagged: they are placed before all timetable event nodes (conceptually at $-\infty$) so that access links can connect to departure events that occur either before or after the desired departure window without violating acyclicity. Centroid-out nodes are not time-tagged and are placed after all event nodes (conceptually at $+\infty$), preserving a global topological order.

Remark. The separation of centroid-in and centroid-out nodes is essential for the validity of the forward dynamic-programming loading algorithm. Without this separation, egress links could break the acyclic structure and violate the topological order implied by time.

Definition 6 (Links)**.** *The link set* $\mathcal{A}$ *contains the following link types:*

1. ***Access links (time-windowed):*** *for an OD pair* $q = (s_o, s_d)$ *and departure interval* t, *access links connect the origin centroid-in node* $(s_o, t)^{\text{in}}$ *to* selected departure *event nodes at the origin stop,*

$$(s_o, t)^{\text{in}} \to (s_o, \tau)^{\text{dep}},$$

   only for departure times τ *satisfying*

$$\tau \in \left[a_t - \Delta^{\text{access}}_{\text{max}},\ b_t + \Delta^{\text{access}}_{\text{max}}\right].$$

2. ***Ride links:*** *for each trip, ride links connect a departure event at the current stop to an arrival event at the next stop,*

$$(s_i, \tau^{\text{dep}}_i)^{\text{dep}} \to (s_{i+1}, \tau^{\text{arr}}_{i+1})^{\text{arr}}.$$

3. ***Dwell/continue links (within-trip at a stop):*** *for each trip and each stop event, a dwell link connects the arrival and departure nodes at the same stop,*

$$(s, \tau^{\text{arr}})^{\text{arr}} \to (s, \tau^{\text{dep}})^{\text{dep}}.$$

   In the implementation, strictly positive dwell time is enforced by regularizing timetable records where arrival equals departure (using a small minimum dwell time).

4. ***Transfer links (inter-line, time-windowed):*** *transfer links connect an* arrival *event node to a later* departure *event node at the same stop and represent waiting and switching* between different lines. *A transfer link*

$$(s, \tau_1)^{\text{arr}} \to (s, \tau_2)^{\text{dep}}, \qquad \textit{with} \qquad \tau_2 > \tau_1$$

   is included only if

   (a) *the two event nodes belong to* different lines, *and*

   (b) *the waiting time satisfies* $\tau_2 - \tau_1 \leqslant \Delta^{\text{transfer}}_{\text{max}}$.

   No transfer links are created between events of the same line. In particular, transfer links are never created for $\tau_2 = \tau_1$.

5. ***Egress links (global graph, destination-gated by masking):*** *the global graph contains egress links from every arrival event node at stop* s *to the centroid-out node* s^{out}*:*

$$(s, \tau)^{\text{arr}} \to s^{\text{out}}.$$

   For a given OD pair $q = (s_o, s_d)$, *only the destination centroid-out node* s_d^{out} *is enabled as a terminal node; all other egress options are disabled by a destination-dependent mask.*

Each link $\ell \in \mathcal{A}$ is associated with:

- a generalized cost c_ℓ,
- a capacity u_ℓ for ride links (stored in the data structure, not yet used in costs).

Assumption 1 (Acyclic time-expanded network)**.** *All links in the time-expanded network are consistent with a strict topological order. All* event-to-event *links strictly increase clock time. In addition, centroid-in nodes* $(s, t)^{\text{in}}$ *are placed* before *all event nodes (conceptually at* $-\infty$*), and centroid-out nodes* s^{out} *are placed* after *all event nodes (conceptually at* $+\infty$*). With this convention, access and egress links always respect the global ordering.*

With the event split, feasible movements follow the pattern

$$(s, t)^{\text{in}} \to (s, \tau)^{\text{dep}} \to (s', \tau')^{\text{arr}} \to (s', \tau'')^{\text{dep}} \to \cdots \to (s_d, \bar{\tau})^{\text{arr}} \to s_d^{\text{out}}.$$

Transfer links satisfy $\tau_2 > \tau_1$ *and dwell links are enforced to be strictly forward in time through a minimum dwell regularization. Therefore the directed graph is acyclic and admits a topological ordering, which allows value functions and flows to be computed using dynamic programming.*

2.5 Generalized Cost

Each passenger seeks to minimize a generalized cost that includes:

- in-vehicle travel time (ride links),
- waiting and transfer time (transfer links) and dwell time at stops (dwell/continue links),
- transfer disutility through coefficient β_{transfer},
- early/late departure penalty relative to the desired departure *interval* (access links; zero penalty within $[a_t, b_t]$).

For a path p, the generalized cost is

$$C(p) = \sum_{\ell \in p} c_\ell.$$

2.5.1 Link-type-specific cost formulation

The generalized cost of each link depends on its type. All coefficients appearing below are fixed user-defined parameters.

Ride links. For a ride link ℓ representing the movement of a vehicle between two stops, connecting a departure event to a subsequent arrival event,

$$(s_i, \tau_i^{\text{dep}})^{\text{dep}} \to (s_{i+1}, \tau_{i+1}^{\text{arr}})^{\text{arr}},$$

the generalized cost is equal to the in-vehicle travel time:

$$c_\ell = \tau_{i+1}^{\text{arr}} - \tau_i^{\text{dep}}.$$

Dwell/continue links. For a dwell/continue link at a stop,

$$(s, \tau^{\text{arr}})^{\text{arr}} \to (s, \tau^{\text{dep}})^{\text{dep}},$$

the generalized cost is the dwell time at the stop:

$$c_\ell = \tau^{\text{dep}} - \tau^{\text{arr}}.$$

Transfer links (inter-line, bounded waiting). Consider a transfer link $\ell : (s, \tau_1)^{\text{arr}} \to (s, \tau_2)^{\text{dep}}$ with $\tau_2 > \tau_1$. Such a link is included only if (i) the two events correspond to *different lines* and (ii) the waiting time $\tau_2 - \tau_1$ does not exceed the user-defined threshold $\Delta_{\text{max}}^{\text{transfer}}$. When included, its generalized cost is

$$c_\ell = \beta_{\text{transfer}} \, (\tau_2 - \tau_1),$$

where $\beta_{\text{transfer}} > 0$ is a user-defined coefficient.

Access links (bounded schedule deviation with a flat region). Consider an access link from the centroid-in node $(s_o, t)^{\text{in}}$ to an origin *departure* event node $(s_o, \tau)^{\text{dep}}$. The link is included only if

$$\tau \in \left[a_t - \Delta_{\text{max}}^{\text{access}},\ b_t + \Delta_{\text{max}}^{\text{access}}\right].$$

When included, its generalized cost is

$$c_\ell = \beta_{\text{early}} \max(0, a_t - \tau) + \beta_{\text{late}} \max(0, \tau - b_t),$$

with user-defined coefficients $\beta_{\text{early}} \geqslant 0$ and $\beta_{\text{late}} \geqslant 0$.

Egress links to centroids. For an egress link $\ell : (s, \tau)^{\text{arr}} \to s^{\text{out}}$, we set $c_\ell = 0$.

Interpretation of cost coefficients. All time quantities in the model are measured in minutes. Travelers may perceive these minutes differently depending on context. The coefficients β_{transfer}, β_{early} and β_{late} act as perception weights converting physical time into generalized cost. These coefficients are fixed and specified by the user; they are not estimated within the inference procedure. Only the OD demand and optionally the dispersion parameter θ are estimated.

3 Assignment Model

3.1 Input

The input is the OD vector $\mathbf{f}$ (scenario order), which is injected through centroid-in nodes.

3.2 Assignment Procedure

The classical assignment procedure based on deterministic shortest paths is not differentiable with respect to link costs and is therefore not well suited for gradient-based calibration methods. We adopt a differentiable assignment model inspired by the logit-based approach of Dial (1971), adapted to the acyclic time-expanded public transportation network.

3.3 Differentiable Dial-style assignment

Let f denote the OD demand. For each OD pair q and departure interval t, passengers choose routes according to a logit model defined on the time-expanded network.

3.3.1 Capacity constraints (current status)

Although ride links are associated with nominal capacities u_ℓ in the data structure, the current implementation does not incorporate capacity-dependent penalties into generalized costs. All assignment computations are therefore performed without congestion or overload effects.

Introducing capacity-aware penalties while preserving differentiability (e.g., using smooth overload functions such as softplus penalties) is a natural future extension, but it is not part of the present version.

3.3.2 Logit routing model

Passengers are assumed to choose routes according to a logit model with dispersion parameter $\theta > 0$. For any path p,

$$\Pr(p) \propto \exp\left(-\frac{C(p)}{\theta}\right), \qquad C(p) = \sum_{\ell \in p} c_\ell.$$

When θ is small, flows concentrate on lowest-cost paths. When θ is large, flows spread across multiple attractive alternatives.

Computational and behavioral motivation. The logit formulation admits an efficient dynamic-programming implementation on the acyclic time-expanded network. It is differentiable with respect to generalized costs and, consequently, with respect to OD demand. This property is essential when the assignment model is embedded in gradient-based calibration and variational inference.

3.3.3 Differentiable dynamic programming formulation

Fix an OD pair $q = (s_o, s_d)$ and a departure interval t. The *terminal* node is the destination centroid-out node s_d^{out}, and the value function is initialized as

$$V(s_d^{\text{out}}) = 0.$$

For any other node i, let $\mathcal{A}_q^+(i)$ denote the set of outgoing links that are enabled for the destination group. We define

$$V(i) = -\theta \log \sum_{\ell=(i\to j)\in \mathcal{A}_q^+(i)} \exp\left(-\frac{c_\ell + V(j)}{\theta}\right).$$

This recursion is evaluated in reverse topological order using a numerically stable `logsumexp` formulation.

The probability that a passenger located at node i chooses link $\ell = (i \to j)$ is

$$P_\ell(i) = \frac{\exp\left(-\frac{c_\ell + V(j)}{\theta}\right)}{\sum_{(i\to k)\in \mathcal{A}_q^+(i)} \exp\left(-\frac{c_{ik} + V(k)}{\theta}\right)}.$$

3.3.4 Flow propagation

Fix an OD pair $q = (s_o, s_d)$ and a departure time interval $t \in \mathcal{T}$. The corresponding demand f_q^t is injected at the origin centroid-in node $(s_o, t)^{in}$.

Let y_i denote the flow reaching node i during the forward pass. Initialization is

$$y_{(s_o,t)^{in}} = f_q^t, \qquad y_i = 0 \quad \text{for all } i \neq (s_o, t)^{in}.$$

Given transition probabilities $P_\ell(i)$ on outgoing links $\ell = (i \to j)$, link flows are

$$x_\ell = y_i \, P_\ell(i), \qquad \ell = (i \to j),$$

and node flows are propagated in topological order by

$$y_j \leftarrow y_j + x_\ell, \qquad \ell = (i \to j).$$

Destination gating. The global graph contains egress links for all stops, but for a given OD pair $q = (s_o, s_d)$ the model enables only egress links leading to s_d^{out} via a destination-dependent mask.

Batched evaluation. For computational efficiency, value functions and flow propagation are computed in batches for OD groups sharing the same destination (and, depending on the implementation, the same departure interval), enabling vectorized evaluation with JAX.

3.3.5 Opt-out alternative (future extension)

In a general formulation, an opt-out alternative can be introduced for each OD pair and departure interval. In the first implementation, opt-out alternatives are not included: all demand is assumed to be assigned to the public transportation network.

3.4 Implementation notes for fast JAX evaluation

The time-expanded network is fixed across evaluations. All graph connectivity data are precomputed once and stored as immutable arrays: a topological ordering of nodes; compressed representations of outgoing adjacency; and, for each link, tail/head indices and link type.

The backward recursion for the value function and forward propagation for flows are implemented using JAX primitives such as `logsumexp` and scan-like passes over the topological order. Variable out-degrees are handled through CSR-like adjacency and/or padded adjacency with masks, enabling compilation with `jit`.

3.4.1 Deterministic indexing requirement

Reproducibility of inference results requires that the indexing of nodes, links, and OD cells remain strictly deterministic across runs. Graph construction, link ordering, and OD indexing must therefore follow fixed ordering conventions (e.g., sorted iteration and deterministic array construction). Any change in indexing invalidates stored inference outputs, as the Bayesian layer relies on a fixed mapping between parameters and link indices.

4 Measurement Model and Likelihood

Let Y denote observed data (e.g., passenger counts on selected links, or aggregates derived from them). Operational measurements are available only for a subset of quantities. Let $\mathcal{M}$ denote the index set of available measurements. For each $m \in \mathcal{M}$ we denote by Y_m the observed count.

Predicted measurement from the assignment model. The assignment model produces link flows $\mathbf{x}(\mathbf{f};\theta) = \{x_\ell(\mathbf{f};\theta) : \ell \in \mathcal{A}\}$. A deterministic mapping associates each measurement m with a subset of links $\mathcal{A}_m \subseteq \mathcal{A}$, so that the model-predicted quantity is the sum of the corresponding link flows:

$$\lambda_m(\mathbf{f};\theta) = \sum_{\ell \in \mathcal{A}_m} x_\ell(\mathbf{f};\theta).$$

In the current implementation, aggregation weights are implicitly equal to one. Allowing general nonnegative weights is a possible extension.

Asymmetric counting device (undercounting). The counting device is asymmetric: it is more likely to *miss* passengers than to count passengers that are not present. We model this by introducing a detection rate $\rho \in (0, 1]$ such that the expected observed count is

$$\mu_m(\mathbf{f};\theta,\rho) = \rho\,\lambda_m(\mathbf{f};\theta).$$

Negative binomial likelihood. Conditionally on $(\mathbf{f}, \theta, \rho, r)$, the observations are assumed independent and

$$Y_m \mid \mathbf{f}, \theta, \rho, r \;\sim\; \mathrm{NegBin}\big(r,\ p_m(\mathbf{f};\theta,\rho,r)\big), \qquad m \in \mathcal{M},$$

where $r > 0$ is a dispersion parameter and the success probability p_m is defined so that $\mathbb{E}[Y_m] = \mu_m(\mathbf{f};\theta,\rho)$:

$$p_m(\mathbf{f};\theta,\rho,r) = \frac{r}{r + \mu_m(\mathbf{f};\theta,\rho)} = \frac{r}{r + \rho\,\lambda_m(\mathbf{f};\theta)}.$$

Hence, for $y \in \mathbb{N}_0$,

$$\Pr(Y_m = y \mid \mathbf{f}, \theta, \rho, r) = \frac{\Gamma(y + r)}{\Gamma(r)\, y!}\,(1 - p_m(\mathbf{f};\theta,\rho,r))^y\,(p_m(\mathbf{f};\theta,\rho,r))^r.$$

Fixed measurement parameters. In the current implementation, the detection rate ρ and the dispersion parameter r are treated as fixed, user-specified hyperparameters and are not estimated within the Bayesian inference procedure. Only the OD demand vector and optionally the dispersion parameter θ are inferred. Joint estimation of (ρ, r) is a possible future extension.

Likelihood and normalization. The likelihood is the product over observed measurements:

$$\mathcal{L}\big(\{Y_m\}_{m\in\mathcal{M}} \mid \mathbf{f}, \theta, \rho, r\big) = \prod_{m\in\mathcal{M}} \Pr\big(Y_m = y_m^{\mathrm{obs}} \mid \mathbf{f}, \theta, \rho, r\big).$$

The corresponding log-likelihood is

$$\ell(\mathbf{f},\theta,\rho,r) = \sum_{m\in\mathcal{M}} \Big[\log\Gamma\big(y_m^{\mathrm{obs}} + r\big) - \log\Gamma(r) - \log\big(y_m^{\mathrm{obs}}!\big) + r\log p_m(\mathbf{f};\theta,\rho,r) + y_m^{\mathrm{obs}}\log(1 - p_m(\mathbf{f};\theta,\rho,r))$$

For posterior computations, terms constant with respect to $(\mathbf{f}, \theta)$ may be dropped.

Alternative observation models (future extension). Other likelihood choices (e.g., Gaussian noise on counts or Poisson models) may be implemented in the future, but the current baseline implementation uses the negative binomial likelihood described above.

5 Bayesian Inference via Variational Inference

Let $\boldsymbol{\delta}$ denote the vector of log-deviations and θ the dispersion parameter of the route choice model. The posterior distribution is proportional to

$$p(\boldsymbol{\delta}, \theta \mid Y) \propto \mathcal{L}(Y \mid \boldsymbol{\delta}, \theta)\, p(\boldsymbol{\delta})\, p(\theta).$$

Exact Bayesian inference is computationally intractable for high-dimensional OD vectors. The current implementation therefore uses *stochastic variational inference* (SVI) as implemented in NumPyro.

Variational approximation. The posterior is approximated by a parametric distribution $q_\phi(\boldsymbol{\delta}, \theta)$. The parameters ϕ are chosen to maximize the *Evidence Lower Bound* (ELBO):

$$\mathrm{ELBO}(\phi) = \mathbb{E}_{q_\phi}\left[\log p(Y, \boldsymbol{\delta}, \theta) - \log q_\phi(\boldsymbol{\delta}, \theta)\right].$$

Maximizing the ELBO is equivalent to minimizing $\mathrm{KL}(q_\phi \,\|\, p)$ between the variational approximation and the true posterior.

Guide family. The baseline implementation uses Gaussian variational families in an unconstrained parameter space (e.g., diagonal or low-rank approximations), and optimization is performed using stochastic gradients (Adam).

Prior on θ. The dispersion parameter θ is strictly positive. In practice, inference is performed in an unconstrained space (e.g., via a log-transformation), and a weakly informative prior is used to avoid degenerate values (e.g., extremely small θ). The exact prior family is configurable in the NumPyro implementation.

Alternative inference engines. Markov chain Monte Carlo methods could be used in principle, but they are computationally expensive in high-dimensional OD settings. Support for alternative probabilistic programming frameworks (e.g., PyMC) may be considered in the future; the present baseline is NumPyro+SVI.

6 Discussion

This framework allows us to combine timetable information and operational counts to infer OD demand and quantify uncertainty, while using a differentiable assignment model compatible with gradient-based inference.

Future extensions include: incorporating capacity effects in a differentiable manner, estimating additional measurement parameters (e.g., ρ and r), supporting weighted measurement aggregations, and adding opt-out alternatives.

References

Dial, R. B. (1971). A probabilistic multipath traffic assignment model which obviates path enumeration, *Transportation research* **5**(2): 83–111.